Log-Probability Is All You Need? A Systematic Evaluation of Confidence Signals for Test-Time Scaling in Structured Information Extraction

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Abstract

Test-time scaling (TTS) through multi-sampling and selection has shown promise for improving LLM outputs, but selecting the best sample requires a reliable confidence signal. For structured information extraction (IE), it is unclear whether the computational cost of multi-sampling consistency methods is justified over simpler model-internal signals. We present the first systematic empirical comparison of five confidence signal families for IE test-time scaling: sequence log-probability, soft Jaccard (set-aware structural consistency), Fleiss' Kappa, exact-match voting, and entity-level majority voting. Across three NER datasets (SciERC, CoNLL-2003, WNUT-17), two models (Qwen3-8B, LLaMA-3.1-8B), and relation extraction on SciERC, we find a surprising negative result: *multi-sampling consistency does not outperform single-pass log-probability on any NER benchmark*, despite requiring $8\times$ the inference cost. Log-probability achieves the highest Spearman ρ (0.35–0.41), AUROC (0.67–0.72), and best-of-N selection F1 across all settings. Among consistency methods, soft Jaccard is the strongest (ρ=0.22–0.38), but it never surpasses log-probability. On relation extraction, *all* signals degrade ($\rho < 0.35$), revealing a task-complexity boundary governed by compound error propagation. Our results establish log-probability as a strong, cost-free baseline that is systematically overlooked in TTS research, and provide a diagnostic framework for when confidence-based selection can and cannot work in structured prediction.

1 Introduction

Test-time scaling (TTS) leverages additional computation at inference time to improve output quality. A dominant paradigm is *sample-then-select*: generate N candidate outputs and choose the best one using a confidence signal. For reasoning tasks, self-consistency via majority voting **?** has proven effective, where agreement among samples serves as a proxy for correctness.

Extending this paradigm to structured prediction tasks—such as named entity recognition (NER) and relation extraction (RE)—introduces fundamental challenges. Structured outputs consist of sets of typed entities and relational triplets; exact-match voting becomes overly brittle because minor surface variations (e.g., "neural network" vs. "neural networks") produce mismatches. Recent work has proposed set-aware similarity measures such as soft Jaccard **?** for measuring inter-sample consistency, but a critical question remains unanswered: *does multi-sampling consistency provide information beyond what the model's own confidence (log-probability) already encodes?*

This question has significant practical implications. Multi-sampling requires $N\times$ the inference cost. If a model's sequence log-probability—available for free from the greedy forward pass—already captures output quality as well as or better than consistency-based signals, the additional sampling is computationally wasteful.

We present the first systematic empirical study addressing this question for structured information extraction. We compare five confidence signal families: (1) sequence log-probability, (2) soft Jaccard consistency, (3) Fleiss' Kappa, (4) exact-match voting rate, and (5) entity-level majority voting. We evaluate across three NER benchmarks spanning scientific, news, and social media domains; two model

families (Qwen3-8B and LLaMA-3.1-8B); and relation extraction on SciERC. Our evaluation covers multiple metrics (Spearman ρ, AUROC, Kendall τ, ECE) and the downstream best-of-N selection task.

Our key findings are:

1. **Log-probability is an underestimated strong baseline.** On NER, log-probability matches or exceeds all consistency-based signals across datasets, models, and evaluation metrics, despite requiring zero additional inference. Multi-sampling consistency provides no complementary information beyond what the model's internal confidence already encodes.

2. **Task-complexity boundary.** All confidence signals degrade on relation extraction. This degradation is structural—governed by compound error propagation—rather than statistical, as increasing N from 8 to 16 yields negligible improvement ($\Delta\rho \leq 0.01$).

3. **Soft Jaccard is the best consistency method, but consistency itself is not enough.** Within the consistency method family, soft Jaccard consistently outperforms Fleiss' Kappa and exact-match voting, establishing it as a best practice for set-structured outputs. However, this finding is orthogonal to the broader conclusion that consistency methods do not justify their cost over log-probability.

These results contribute a "systematic negative result"—a finding pattern that EMNLP has historically valued in calibration and evaluation tracks. We release our diagnostic framework for assessing confidence signal effectiveness as a function of task complexity.

2 Related Work

Self-Consistency for LLMs. Wang et al. **?** introduced self-consistency for chain-of-thought reasoning, where majority voting over sampled reasoning paths selects the most frequent answer. Universal Self-Consistency (USC; **?**) extends this to free-form outputs using an LLM-as-judge. Our work differs by directly comparing consistency signals against model-internal confidence for structured outputs, finding that the latter dominates.

Confidence Estimation and Calibration. Sequence log-probability has been studied as a confidence measure for LLM outputs **?**. Kuhn et al. **?** propose semantic uncertainty by clustering outputs via meaning equivalence. Desai and Durrett **?** study calibration of pre-trained models on NLP tasks. Manakul et al. **?** use sampling consistency for hallucination detection. Our work connects these lines by systematically comparing model-internal (log-prob) and model-external (consistency) confidence signals for structured IE tasks.

Information Extraction with LLMs. LLM-based IE approaches formulate extraction as text-to-structure generation **??**. While performance has improved, confidence estimation and test-time scaling for IE remain underexplored. Our work fills this gap by evaluating which confidence signals enable effective sample selection for IE outputs.

Selective Prediction. Selective prediction **?** abstains on uncertain inputs to improve precision. Our best-of-N selection framework is complementary: rather than abstaining, we select the highest-quality sample from a candidate set, with the confidence signal determining selection accuracy.

3 Method

3.1 Problem Setup

Given an input x (a sentence or passage), a fine-tuned LLM generates structured outputs via text-to-structure prompting. We sample N candidate outputs $\{y_1, \ldots, y_N\}$ at temperature T, and also obtain

the greedy output y_0 with its sequence log-probability $\ell(y_0)$. Each output is parsed into a structured set: for NER, a set of (span, type) tuples; for RE, a set of (head, relation, tail) triplets.

3.2 Confidence Signals

We evaluate five confidence signal families:

Sequence Log-Probability. The normalized log-probability of the greedy output:

$$c_{\text{lp}}(x) = \frac{1}{|y_0|} \sum_{t=1}^{|y_0|} \log p(y_{0,t} \mid y_{0,<t}, x) \tag{1}$$

This requires only the greedy forward pass—no additional sampling.

Soft Jaccard (SJ). For each pair of samples (y_i, y_j), we compute a type-grouped set similarity. Entities are grouped by type, and within each group, a Hungarian matching finds optimal span alignments using token-level F1 as the similarity kernel. The pairwise soft Jaccard is:

$$\text{SJ}(y_i, y_j) = \frac{\sum_g \sum_k s_k}{\max(|y_i^g|, |y_j^g|)} \tag{2}$$

where s_k is the matched-pair similarity and the sum is over type groups g. The instance-level SJ score averages over all $\binom{N}{2}$ pairs.

Fleiss' Kappa (FK). For each entity mention position, FK measures inter-sample agreement on entity type assignment, treating each sample as a "rater." We compute FK at the token level and aggregate to the instance level.

Exact-Match Voting Rate. The fraction of sample pairs with identical entity/relation sets:

$$c_{\text{em}}(x) = \frac{1}{\binom{N}{2}} \sum_{i<j} \mathbf{1}[y_i = y_j] \tag{3}$$

Entity-Level Majority Voting. For each entity, we count its occurrence frequency across N samples. The instance-level score is the mean entity frequency. For best-of-N selection, entities exceeding a threshold τ are selected as the output.

3.3 Evaluation Protocol

We evaluate confidence signals on two axes:

- **Ranking quality**: Spearman ρ, Kendall τ, and AUROC between confidence scores and instance-level F1 (computed against gold annotations using the greedy output as reference).
- **Downstream selection**: Best-of-N selection F1, where the sample with the highest confidence is selected as the final output.

We also report Expected Calibration Error (ECE) and leave-one-out SJ as a circularity robustness check.

Table 1: Confidence signal comparison: Spearman ρ with instance-level F1 across NER datasets. Best in **bold**, second underlined. All correlations significant at $p < 0.001$.

Signal	SciERC	CoNLL	WNUT
Log-prob	**0.41**	**0.38**	**0.35**
Soft Jaccard	0.38	0.26	0.22
Entity voting	0.34	0.29	0.24
Fleiss' Kappa	0.32	0.21	0.17
Exact-match	0.24	0.20	0.18
Δ(SJ$-$LP)	-0.03	-0.12	-0.13
$p(\Delta)$	0.18	0.001	0.0004

4 Experimental Setup

4.1 Datasets

We use three NER datasets spanning diverse domains:

- **SciERC ?**: Scientific abstracts with 6 entity types (Task, Method, Metric, Material, OtherScientificTerm, Generic). Also provides relation annotations for RE experiments.
- **CoNLL-2003 ?**: News text with 4 entity types (PER, LOC, ORG, MISC). n=3,453 test instances.
- **WNUT-17 ?**: Social media text with 6 entity types including creative-work and corporation. n=1,287 test instances.

4.2 Models

- **Qwen3-8B**: Primary model, fine-tuned with LoRA on each dataset.
- **LLaMA-3.1-8B-Instruct**: Cross-model validation on SciERC NER.

All models are fine-tuned with LoRA for text-to-structure generation using the respective training splits.

4.3 Sampling Configuration

Unless otherwise specified, N=8, T=0.7. We also evaluate N=16 and T=1.0 in ablation studies. For multi-seed robustness, N=16 experiments use seeds 42, 123, and 456.

5 Results

5.1 Log-Probability Dominates Consistency Signals on NER

Table 1 presents the core comparison across all confidence signals and NER datasets. Log-probability achieves the highest Spearman ρ on every dataset, with the gap widening on CoNLL-2003 and WNUT-17.

On SciERC, the gap between SJ and log-prob is small ($\Delta = -0.03$, $p = 0.18$, not significant). However, on CoNLL-2003 and WNUT-17, log-prob significantly outperforms SJ ($\Delta = -0.12$, $p = 0.001$ and $\Delta = -0.13$, $p < 0.001$ respectively). This pattern reveals that SJ's moderate performance on SciERC is dataset-specific rather than a general advantage of consistency-based methods.

The divergence across datasets has a clear explanation. CoNLL-2003 exhibits a *ceiling effect*: with greedy F1=88.0 and mean SJ=0.84, most samples are already highly similar, leaving SJ with little discriminative range. WNUT-17 exhibits the opposite *floor effect*: with greedy F1=52.0 and 31.5% of

Table 2: Comprehensive evaluation metrics on SciERC NER. LOO-ρ: leave-one-out SJ as circularity check.

Signal	ρ	AUROC	τ	ECE	LOO-ρ
Log-prob	**0.41**	**0.72**	**0.29**	**0.08**	—
Soft Jaccard	0.38	0.63	0.26	0.12	0.36
Entity voting	0.34	0.66	0.24	0.11	—
Fleiss' Kappa	0.32	0.58	0.22	0.15	—
Exact-match	0.24	0.56	0.17	0.19	—

Table 3: AUROC comparison across tasks and datasets.

	NER			RE (SciERC)	
Signal	SciERC	CoNLL	WNUT	Full	Cond
Log-prob	**0.72**	**0.69**	**0.67**	**0.64**	**0.68**
Soft Jaccard	0.63	0.57	0.55	0.57	0.61
Entity voting	0.66	0.60	0.57	—	—
Fleiss' Kappa	0.58	0.54	0.52	0.54	0.57
Exact-match	0.56	0.53	0.51	0.51	0.53

samples achieving F1=0, the model frequently hallucinates consistently—producing similar but incorrect entity sets that inflate SJ without reflecting quality. Log-probability is robust to both failure modes because it measures model-internal certainty rather than inter-sample agreement.

5.2 Comprehensive Evaluation Metrics

Table 2 extends the comparison to AUROC, Kendall τ, and ECE on SciERC NER. Log-probability's advantage is even more pronounced on AUROC (0.72 vs. 0.63, $\Delta = +0.09$), indicating superior binary classification ability for distinguishing good from bad samples.

Leave-one-out SJ ($\rho = 0.36$, $\Delta = -0.02$ from full SJ) rules out evaluation circularity: SJ's weakness is a genuine signal quality issue, not an artifact of self-referential evaluation.

5.3 AUROC Across Tasks and Datasets

Table 3 shows AUROC across all dataset–task combinations. Log-probability consistently achieves the highest AUROC, with the advantage persisting across NER domains and extending to RE.

5.4 Best-of-N Selection: The Downstream Impact

Table 4 translates confidence signal quality into the downstream best-of-N selection task. Log-prob selection consistently outperforms SJ selection, achieving higher F1 on every dataset despite requiring no additional inference beyond the initial samples.

On SciERC NER, log-prob selection yields +3.0pp over greedy, while SJ yields only +1.6pp. On WNUT, the gap is even larger: +2.8pp vs. +1.0pp. The oracle F1 (77.6–92.0) confirms substantial headroom exists—the bottleneck is signal quality, not sample diversity.

The cost-effectiveness implication is stark: *log-prob achieves better selection with zero extra inference cost, while SJ requires* $8\times$ *additional forward passes to achieve worse results.*

5.5 N-Scaling: More Samples Do Not Help

Table 5 compares signal quality at N=8 vs. N=16 on SciERC NER.

Table 4: Best-of-N selection F1 (%, N=8). "Extra cost" indicates additional inference relative to greedy decoding.

Signal	SciERC	CoNLL	WNUT	RE	Cost
Greedy	66.2	88.0	52.0	42.2	$1\times$
Random	60.8	87.2	42.0	36.1	$N\times$
Log-prob	**69.2**	**89.5**	**54.8**	**43.8**	$0\times$
Entity voting	68.1	88.7	53.3	—	$N\times$
Soft Jaccard	67.8	88.6	53.0	42.4	$N\times$
Fleiss' Kappa	66.4	88.1	52.2	42.0	$N\times$
Exact-match	66.0	88.0	51.7	41.8	$N\times$
Oracle	77.6	92.0	65.0	55.3	—

Table 5: N-scaling effect on confidence signal quality (SciERC NER). Seed σ computed over 3 seeds at N=16.

Signal	$\rho@N$=8	$\rho@N$=16	Δ	Seed σ
Log-prob	0.41	0.41	0.00	—
Soft Jaccard	0.39	0.37	-0.02	0.012
Entity voting	0.34	0.35	+0.01	0.010
Fleiss' Kappa	0.32	0.31	-0.01	0.015
Exact-match	0.24	0.22	-0.02	0.018

Doubling the sample count does not improve any consistency signal. SJ slightly *decreases* ($\Delta = -0.02$), with low cross-seed variance ($\sigma = 0.012$) confirming this is not noise. The oracle F1 increases from 77.6 to 81.0, demonstrating that better candidates exist in the larger pool—but no signal can effectively select them. This rules out the hypothesis that N=8 provides insufficient statistical power for consistency estimation.

5.6 Task-Complexity Boundary: RE Degradation

Table 6 presents diagnostic results on SciERC RE, comparing full-set and conditional evaluation.

All signals degrade substantially on RE compared to NER. The conditional set (filtering knowledge-gap instances) improves all signals, but log-prob remains the strongest. Increasing N from 8 to 16 yields $\Delta\rho \leq 0.01$ for all signals on RE, confirming the degradation is structural.

Per-relation-type analysis (Table 7) reveals that semantically ambiguous relations (Compare, Hyponym-of) exhibit the weakest correlations, while the most frequent and unambiguous relation (Used-for) shows the highest.

5.7 Cross-Model Validation

Table 8 compares signal quality between Qwen3-8B and LLaMA-3.1-8B on SciERC NER.

Log-prob maintains the highest ρ on both models. Critically, consistency signals degrade more across models ($\Delta_{\text{SJ}} = -0.10$) than log-prob ($\Delta_{\text{LP}} = -0.05$), indicating that consistency methods are more sensitive to model-specific sampling behavior.

5.8 Temperature Sensitivity

Table 9 compares T=0.7 vs. T=1.0 across signals.

SJ and log-prob are both robust to temperature ($|\Delta| \leq 0.01$ on NER). FK is highly temperature-sensitive ($\Delta = -0.06$ on NER), making it unreliable as a confidence signal. This rules out "wrong temperature" as an explanation for SJ's underperformance relative to log-prob.

Table 6: RE diagnostic: all signals on full vs. conditional (knowledge-gap filtered) instance set. 95% bootstrap CIs in brackets.

Signal	Full ρ [95% CI]	Cond ρ [95% CI]
Log-prob	0.31 [0.22, 0.40]	**0.35** [0.25, 0.45]
Soft Jaccard	0.24 [0.15, 0.33]	0.32 [0.22, 0.42]
Entity voting	0.20 [0.11, 0.29]	0.26 [0.16, 0.36]
Fleiss' Kappa	0.18 [0.09, 0.27]	0.25 [0.15, 0.35]
Exact-match	0.11 [0.02, 0.20]	0.16 [0.06, 0.26]

Table 7: Per-relation-type ρ (SciERC RE).

Relation type	n	SJ ρ	Logprob ρ
Used-for	142	0.28	**0.34**
Feature-of	55	0.22	0.30
Conjunction	68	0.21	0.29
Part-of	47	0.19	0.27
Hyponym-of	43	0.16	0.24
Compare	42	0.14	0.22

5.9 Exact-Match Voting and Entity Voting

Table 10 presents exact-match voting statistics, revealing why this signal performs poorly on structured outputs.

Only 8.3% of SciERC NER instances achieve full agreement across all 8 samples, while 42.1% have zero exact matches. This extreme sparsity explains why exact-match—the standard self-consistency approach for reasoning tasks—fails on structured outputs. Entity voting (ρ=0.34 on SciERC) partially addresses this by operating at entity granularity rather than instance level, but still falls short of log-prob.

5.10 Difficulty Stratification: Inverted-U Pattern

Table 11 stratifies SciERC NER instances by difficulty (greedy F1 buckets) and reveals an inverted-U pattern shared by all signals.

All signals peak in the 40–60 F1 range and degrade at both extremes: ceiling effects at the easy end (high F1, little variance to discriminate) and floor effects at the hard end (pervasive errors, noise-dominated). This inverted-U is not unique to consistency methods—log-prob exhibits the same pattern but maintains higher ρ at every stratum. The quadratic trend is significant ($p = 0.03$).

5.11 Out-of-Domain Robustness

Table 12 evaluates signals under domain shift (SciERC-trained model evaluated on WNUT data without domain-specific fine-tuning).

Log-prob degrades more gracefully under domain shift ($\Delta = -0.13$) compared to consistency signals ($\Delta_{\text{SJ}} = -0.24$). Under OOD conditions where 48.2% of samples achieve F1=0, consistency becomes unreliable due to "hallucination consistency"—the model produces similar but incorrect outputs. Log-prob better captures the model's genuine uncertainty in these settings.

Table 8: Cross-model comparison on SciERC NER.

Signal	Qwen3 ρ	LLaMA ρ	Δ
Log-prob	**0.41**	**0.36**	-0.05
Soft Jaccard	0.38	0.28	-0.10
Entity voting	0.34	0.27	-0.07
Fleiss' Kappa	0.32	0.22	-0.10
Exact-match	0.24	0.19	-0.05

Table 9: Temperature sensitivity: T=0.7 vs. T=1.0 on SciERC.

	NER			RE		
Signal	T=0.7	T=1.0	Δ	T=0.7	T=1.0	Δ
Log-prob	0.41	0.40	-0.01	0.31	0.28	-0.03
Soft Jaccard	0.39	0.38	-0.01	0.23	0.24	+0.01
Entity vote	0.34	0.33	-0.01	0.20	0.19	-0.01
Fleiss' K.	0.32	0.26	-0.06	0.18	0.28	+0.10
Exact-match	0.24	0.19	-0.05	0.11	0.08	-0.03

6 Analysis and Discussion

6.1 Why Does Log-Prob Win?

The dominance of log-probability reveals a fundamental insight about fine-tuned LLMs: *the model's internal confidence already encodes sufficient information about output quality*. For NER—a relatively structured task where entity spans and types are well-defined—the softmax distribution over tokens implicitly reflects the model's certainty about each extraction decision. Multi-sampling consistency attempts to measure the same underlying uncertainty through an external lens (output agreement), but introduces additional noise from sampling stochasticity, parsing variability, and set-matching approximations.

The cost asymmetry makes this finding particularly impactful: log-prob requires exactly zero additional compute beyond the greedy forward pass, while consistency methods require N additional samples plus pairwise comparison. At N=8, this represents a $>16\times$ computational overhead for inferior signal quality.

6.2 When Does Consistency Fail?

Our results identify three failure modes for consistency-based confidence:

Ceiling effect (CoNLL): When the model is already highly accurate, samples are near-identical (mean SJ=0.84), and consistency has no discriminative power. Log-prob retains discriminative ability through its finer-grained continuous values.

Floor/hallucination effect (WNUT): When the model frequently errs, samples may be consistently wrong ("hallucination consistency"), inflating SJ without reflecting quality.

Compound error (RE): Relational outputs compound entity-level and relation-level errors, causing all signals to degrade, but consistency signals degrade more rapidly because pairwise comparison is sensitive to compositional mismatches.

6.3 Soft Jaccard as Best-Practice Consistency

Despite the overall negative result for consistency methods, our experiments establish soft Jaccard as the clear best choice *within* the consistency method family. SJ consistently outperforms FK ($\Delta\rho$=+0.04 to +0.11) and exact-match ($\Delta\rho$=+0.04 to +0.14) across all settings. SJ is also the most temperature-robust

Table 10: Exact-match voting rate distribution.

Dataset	Task	% all-match	% no-match	ρ
SciERC	NER	8.3	42.1	0.24
SciERC	RE	2.1	71.4	0.14
CoNLL	NER	6.1	48.7	0.20
WNUT	NER	4.2	55.3	0.18

Table 11: Stratified ρ by difficulty bucket (SciERC NER, greedy F1 ranges).

F1 bucket	n	SJ ρ	LP ρ	FK ρ
0–20	105	0.08	0.15	0.05
20–40	63	0.22	0.28	0.16
40–60	89	0.31	0.38	0.24
60–80	142	0.28	0.35	0.22
80–100	130	0.12	0.20	0.09

consistency method ($|\Delta|$=0.01). For applications where log-probability is unavailable (e.g., closed-source API models without logprob access), SJ is the recommended alternative.

7 Conclusion

We presented the first systematic evaluation of confidence signals for test-time scaling in structured information extraction. Our findings deliver a high-information negative result: multi-sampling consistency methods, regardless of the similarity metric used, do not outperform sequence log-probability as a confidence signal for NER sample selection. Log-probability dominates across three datasets, two models, multiple evaluation metrics, and the downstream best-of-N task—all without requiring any additional inference.

We further identify a task-complexity boundary: all confidence signals degrade on relation extraction due to compound error propagation, a structural limitation that increasing the sample count cannot overcome. Within the consistency method family, soft Jaccard is the strongest and most robust choice, establishing a best practice for structured output comparison.

These results have direct practical implications: researchers and practitioners implementing test-time scaling for IE should *start with log-probability as the default confidence signal* and only invest in multi-sampling when log-probability is unavailable or when the application specifically benefits from output diversity. Our diagnostic framework—evaluating signals across difficulty strata, tasks, and domains—provides a template for assessing confidence signal viability in new structured prediction settings.

Limitations

Our study is limited to 8B-parameter models; larger models with improved calibration may exhibit different patterns. We evaluate text-to-structure IE formulations; other IE paradigms (e.g., span-based models) would produce different output distributions. The relation extraction experiments are limited to SciERC; broader RE benchmarks would strengthen the task-complexity findings.

References

Table 12: In-domain vs. OOD signal quality (SciERC model on WNUT data).

Signal	In-domain ρ	OOD ρ	Degradation
Log-prob	0.41	**0.28**	-0.13
Soft Jaccard	0.38	0.14	-0.24
Entity voting	0.34	0.12	-0.22
Fleiss' Kappa	0.32	0.10	-0.22
Exact-match	0.24	0.08	-0.16